

SEAT No. _____

[52]

SARDAR PATEL UNIVERSITY
M.Sc.(IT) Semester-I Examination
PS01CINT53: Computer Networks & Cyber Security
Date : 25th November, 2024

Total No. of printed pages : 2



Time: 10:00 a.m. to 1:00 p.m.

Total Marks: 70

1. Select the most appropriate option for each of the following questions : 8

- (i) If there are 10 nodes in a mesh network, how many links are required ?
(A) 9 (B) 10 (C) 45 (D) None of these.
- (ii) Which of the following networks covers a city ?
(A) LAN (B) MAN (C) WAN (D) None of these.
- (iii) The OSI reference model has _____ layers.
(A) 4 (B) 5 (C) 6 (D) 7
- (iv) _____ is a static routing algorithm in which every incoming packet is sent out on every outgoing line except the one it arrived on.
(A) Flooding (B) Shortest path (C) Link state (D) None of these.
- (v) When two or more networks are connected, they form _____.
(A) an internetwork (B) a LAN (C) an extranet (D) None of these.
- (vi) In _____ fragmentation technique, recombination occurs only at the destination host.
(A) transparent (B) non-transparent (C) semi-transparent (D) None of these.
- (vii) The _____ use the same key for encryption and decryption.
(A) public-key algorithms (B) symmetric-key algorithms
(C) private-key algorithms (D) None of these.
- (viii) The _____ satellites are close to earth and hence zip into and out of view quickly.
(A) LEO (B) MEO (C) Geostationary (D) None of these.

2. Answer the following questions in brief (ANY SEVEN) : 14

- (i) Distinguish between LANs and WANs.
- (ii) Write the functions performed by a **repeater** and an **amplifier**.
- (iii) Specify examples of principal **satellite bands**. What do you mean by a **footprint** of a satellite ?
- (iv) What do you mean by **flooding** ? What are the **uses** of flooding ?
- (v) List the names of various **routing algorithms**.
- (vi) List the names of the **layers** of the **TCP/IP reference model**.
- (vii) What do you mean by a **VPN** ?
- (viii) Define the terms : **plaintext** and **ciphertext**.
- (ix) Explain the meaning of network **congestion**.

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(P.T.O.)

- 3.(A) What do you understand by a **LAN** ? List the salient **characteristics** of LANs. Classify LANs based on **topologies**. 6
- (B) List the layers of the **OSI Reference Model**. Describe the functions performed by **any two** layers. 6
- OR**
- (B) List the **guided transmission media**. Describe the **properties** of **any one** guided transmission medium in detail. 6
- 4.(A) Write a short note on **satellite communication**. List the advantages of satellite communication over fiber optics communication. 6
- (B) What do you mean by a dynamic routing algorithm ? Describe the **distance vector routing algorithm**. What is its **drawback** ? 6
- OR**
- (B) What is meant by static routing ? Describe the **shortest path routing algorithm**. 6
- 5.(A) Explain the term **internetworking**. How networks differ ? List and explain **various issues** that arise when we establish an internetwork. 6
- (B) What is the purpose of using **fragmentation** ? Distinguish between the **transparent fragmentation** and **non-transparent fragmentation** strategies. 6
- OR**
- (B) What do you mean by **tunneling** ? Explain how tunneling works giving a diagram. 6
- 6.(A) What is **cryptography**? What are the fundamental cryptographic principles? What do you understand by symmetric-key algorithms ? 6
- (B) What is the purpose of using a **firewall** ? Explain the typical **configuration** and **working** of a firewall. 6
- OR**
- (B) Explain the terms : **phishing, key loggers** and **password cracking**. 6
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